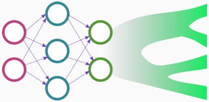
Likelihood-free phylogenetics



Luc Blassel

Auditions CNRS 51/02

May 20th 2026

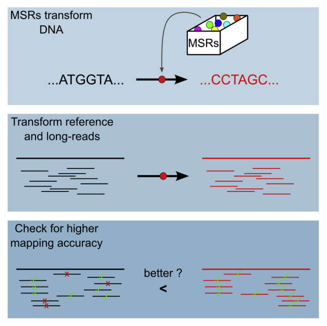
slides: `lucblassel.com/CNRS_2026.pdf`

Professional history

1. Ingénieur @ **AgroParisTech** *2014-2018*
   - Specialisation in **ML** and **data-science**
   - Double diploma with the **IASD** MSc. @ **Dauphine**
2. PhD. @ **Institut Pasteur** *2018-2022*
   (5 papers)
   - *R. Chikhi*: **Improving** sequence **alignements**
   - *O. Gascuel*: **Learning** from **alignments**
3. Post-doc @ **LBBE** & **CQSB** *2023-*
   (1 paper & 2 preprints)
   - *L. Jacob*: **Likelihood-free** phylogenetic **inference**

Previous works - PhD

keywords: Sequence Bioinformatics, combinatorics

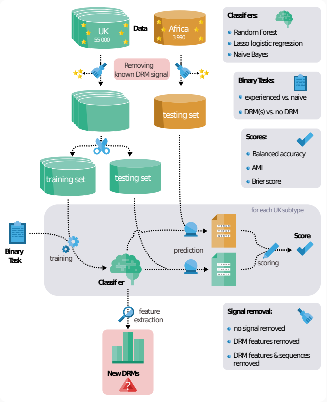


- **Goal**: Improve mapping of NanoPore reads to ref. genome
- **Implementation**: Go programs & Nextflow pipeline
- **Results**: Identified functions that work accross $\neq$ references
- **Impact**: Used for compression in NanoPore assembler[1]

Blassel *et al.* 2022, RECOMB-SEQ

[1] Faure et al. 2025

keywords: Data Analysis, Pipelining, Machine Learning



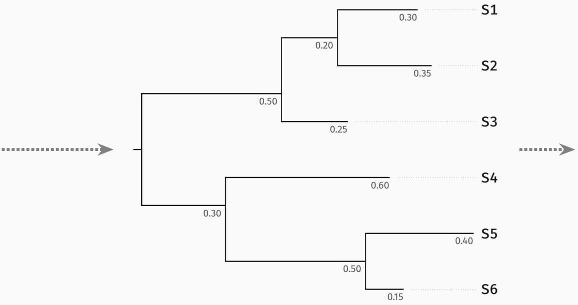
- **Goal**: Find new drug resistance mutations in HIV
- **Method**: Train interpretable classifiers on *treated/non-treated*
- **Implementation**: Python scripts & Snakemake pipeline
- **Results**: Identified new mutations

Blassel *et al.* 2021 PLoS CB & 2021 Curr. Op. in Vir

Research project - Likelihood-free phylogenetics

Project - Context, phylogenetics

S1 TQPRLPTC
S2 TQPSVPKC
S3 TGVPVPAC
S4 TLGRSPSC
S5 TQPRAKTC
S6 TQPRIPAC



- R_0?
- Speciation rate?
- Root age?
- ...

Foundational task in **bioinformatics**

<u>Citations[1]:</u>

NJ: 77k, *PhyML*: 21k+20k, *FastME*: 20k, *IQTree*: 27k+16k, *RaxML*: 36k, *FastTree*: 16k+6k

[1] From Google scholar

Project - Context, phylogenetic inference

x: **sequence** alignment θ: **tree** M: sequence **evolution model**

Inference:

Max. Likelihood: $\theta^* = \underset{\theta}{\operatorname{argmax}}\ \mathbf{p(x|\theta, M)}$

Bayesian: $p(\theta|x, M) \propto \mathbf{p(x|\theta, M)} p(\theta) p(M)$

Problem:

Relies on **repeated** estimations of **Likelihood**:

1. **Costly** and slow **computation**: $\mathcal{O}(ns^3 + nLs^2)$[1]
2. **Limits** model **realism & complexity**

n: nb. of leaves, L: length of sequence, s: size of alphabet

[1]Bryant et al. 2005

Project - Overview

Goal:

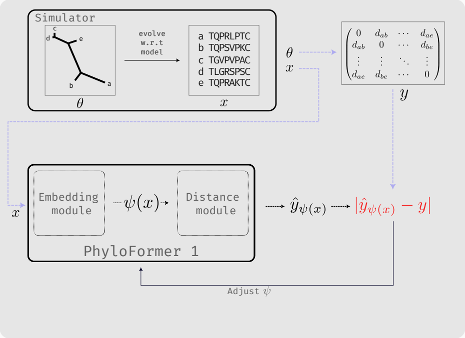
Develop **likelihood-free** phylogenetic **methods**
Enable the use of **richer**, **more realistic** models for phylogenetic tasks

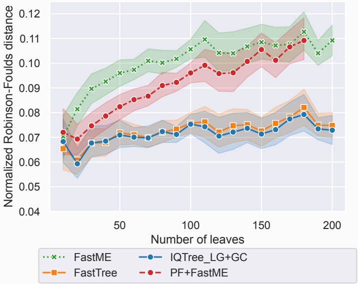
Roadmap for this talk:

1) Initial work ✓	**2) Extending NPE**	**3) Joint inference**
Phyloformer 1 & 2	example: Alignment-free	example: ASR[1]

[1] Ancestral Sequence Reconstruction

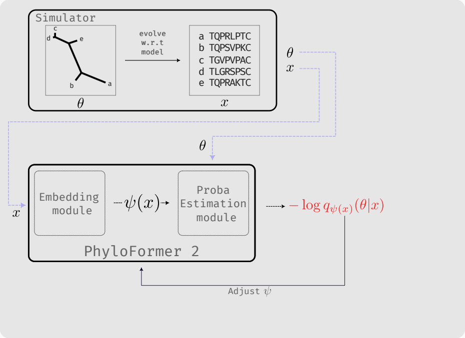
Research Project - initial attempts, Post-doc

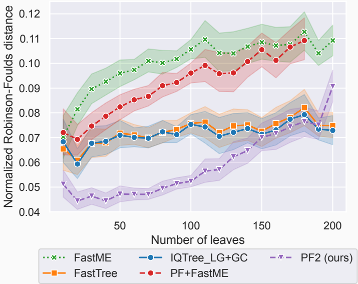




- Use **distance-matrix** as tree **proxy**
- **Inference**: use **existing** distance **methods** on $\hat{y}_{\psi(x)}$

Nesterenko*, **Blassel*** *et al.* 2025, MBE





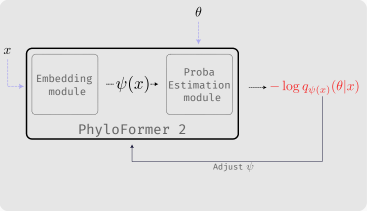
- **Training**: $\psi^* = \underset{\psi}{\text{argmin}} \; -\sum_{i \in train} \log q_{\psi(x_i)}(\theta_i|x_i)$
- **Inference**: we **sample** from $q_{\psi^*(x_E)}(\theta_E|x_E)$

Blassel *et al.* 2025, *preprint*

Research Project - Extending NPE to alignment-free

Problem

- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can lead to **different trees** and **conclusions**[1]

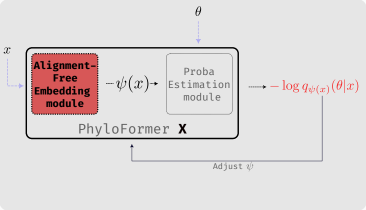


- **PF2** results are **promising**
- **High** potential **impact**

[1] Liu et al. 2011; Wong et al. 2008

Problem

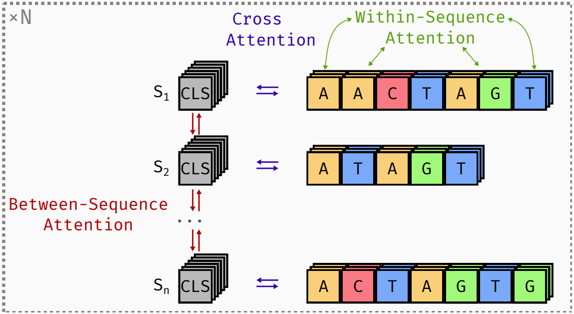
- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can lead to **different trees** and **conclusions**[1]



- **PF2** results are **promising**
- **High** potential **impact**
- "Just" need a **new Embedding** method

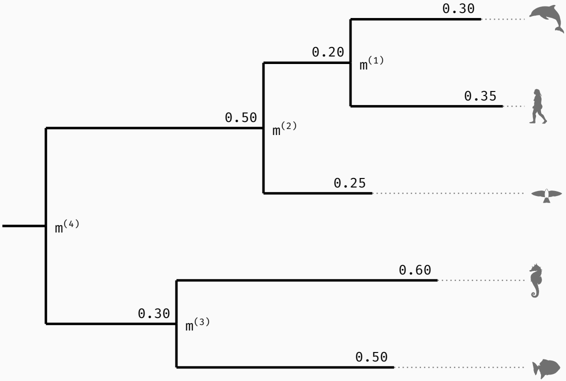
[1] Liu et al. 2011; Wong et al. 2008

Project - An alignment-free embedder



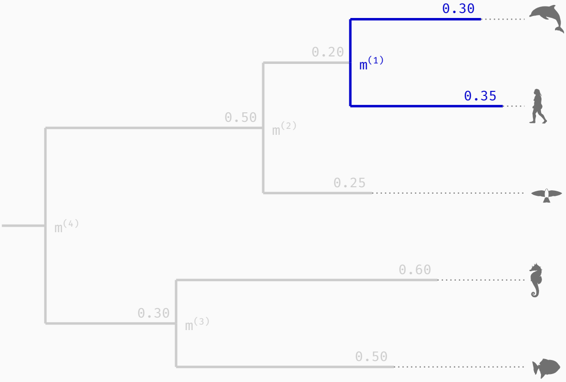
compute $q_{\psi(x)}(\theta|x)$ with $\psi(x) = \{CLS_1, CLS_2, \cdots, CLS_n\}$

Project - Estimating the Posterior



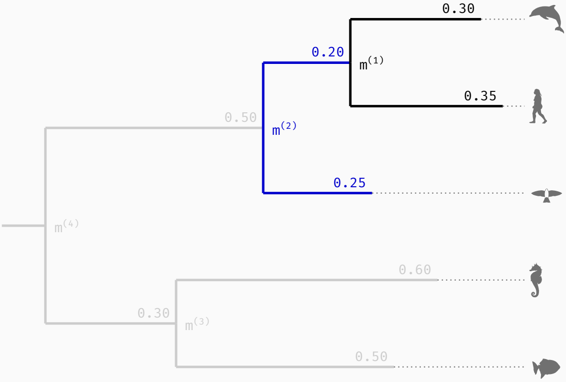
$$q_{\psi(x)}(\theta|x) = ?$$

Project - Estimating the Posterior



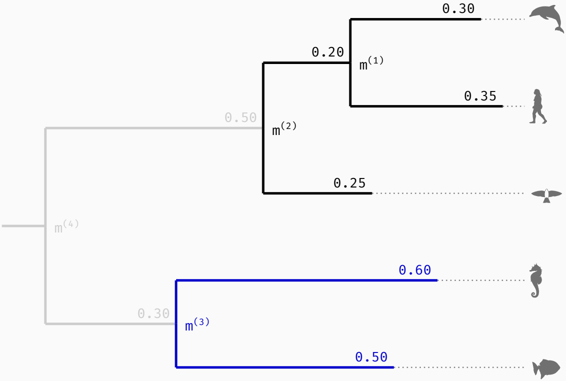
$$q_{\psi(x)}(\theta|x) = \quad p(m^{(1)})$$

Project - Estimating the Posterior



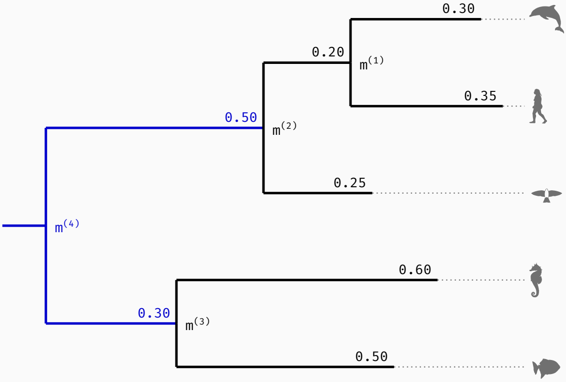
$$q_{\psi(x)}(\theta|x) = \;\; p(m^{(1)}) \times p(m^{(2)}|m^{(1)})$$

Project - Estimating the Posterior



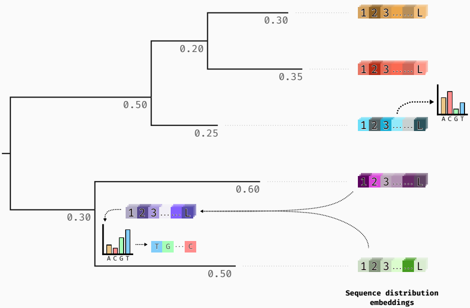
$$q_{\psi(x)}(\theta|x) = \quad p(m^{(1)}) \times p(m^{(2)}|m^{(1)}) \times p(m^{(3)}|m^{(1,2)})$$

Project - Estimating the Posterior



$$q_{\psi(x)}(\theta|x) = \; p(m^{(1)}) \times p(m^{(2)}|m^{(1)}) \times p(m^{(3)}|m^{(1,2)}) \times p(m^{(4)}|m^{(1,2,3)})$$

Project - Joint Tree/Parameter learning, ASR



- **per-site distribution** over alphabet
- *Training*: we have **simulated ancestral** seq.
- *Inference*: we **sample** a seq. at **each node**

Projet - Perspectives

- **Passage à l'échelle**:
  - Appliquer ces méthodes sur $10^{3,4,5\dots}$ séquences.
  - Remplacer l'auto-attention par des approches de meilleur complexité
- **Inférence de réseaux phylogénétiques**:
  - Pour pouvoir modéliser les transferts horizontaux ou la recombination les arbes ne sont pas suffisants
  - L'inférence de réseaux est beaucoup plus compliqué
  - Inférence sans vraisemblance aurait un gros impact sur l'utilisation pratique des réseaux

Intégration

CQSB:

Expertise: ML (L. Jacob), Évolution (M. Weigt, A. Carbone)

Collborations: L. Jacob & P. Barrat-Charlaix

Équipe **Bioinfo** au **LISN**:

Expertise: SBI (F. Pouyet, F. Jay), Arbres & Graphes (L. Planche)

Équipe **Modelling of Pathogens** à l'**IBENS** & **IP**:

Expertise: Évolution & Phylogénie (A. Zhukova, H. Morlon)

Collborations: A. Zhukova

References

Bryant, D. et al. (2005). ***Likelihood Calculation in Molecular Phylogenetics.*** In: *Mathematics of Evolution and Phylogeny*. Ed. by O. Gascuel. Oxford University PressOxford, pp. 33–62.

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Wong, K. M. et al. (2008). ***Alignment Uncertainty and Genomic Analysis.*** In: *Science* 319.5862, pp. 473–476.

Personal Publications

Lemoine, F. et al. (2020). ***COVID-Align: Accurate Online Alignment of hCoV-19 Genomes Using a Profile HMM*.** In: *Bioinformatics* 37.12, pp. 1761–1762.

Blassel, L., A. Tostevin, et al. (2021). ***Using Machine Learning and Big Data to Explore the Drug Resistance Landscape in HIV*.** In: *PLOS Computational Biology* 17.8, e1008873.

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Nesterenko, L. et al. (2025). ***Phyloformer: Fast, Accurate, and Versatile Phylogenetic Reconstruction with Deep Neural Networks*.** In: *Molecular Biology and Evolution* 42.4, msaf051.

Garot, V. et al. (2026). ***Teddy: Neural Inference of Epidemiological Parameters from Viral Sequences*.**

Supplementary slides

Project - Assembly-free phylogenetic inference

Schema

Project - Tree-sequence inference

frame

frame

Project - Joint tree-parameter inference, dating

env